

Innovative Mathematics Olympiad
Model Paper 2025 – 2026

Total Questions: 50

Time: 1.5 hr.

Total Marks: 100

Syllabus

Number System, Polynomials, Coordinate Geometry, Linear Equations in Two Variables, Introduction to Euclid's Geometry, Lines and Angles, Triangles, Quadrilaterals, Circles, Heron's Formula, Surface Areas, and Volumes, Statistics.

Instructions:

1. This question paper contains 50 questions. All questions are compulsory.
2. The duration of the examination is 90 minutes.
3. Each question carries 2 marks. There is no negative marks for wrong answers.
4. Use only blue or black ball-point pen to mark your answers on the OMR Answer Sheet.
5. The use of calculators, mobile phones, smart watches, log tables, or any electronic gadgets is strictly prohibited.

Mathematics

1. The volume of a cuboid of length 10 cm, breadth 5 cm and height 4 cm is _____.
 (A) 100 cm^3 (B) 150 cm^3
 (C) 200 cm^3 (D) 250 cm^3
2. A circular park has a radius of 7 m. What is the distance covered by a person walking once around the park?
 (A) 22 m (B) 44 m
 (C) 49 m (D) 154 m
3. A point P is located at (4, -3). What is the distance of point P from the X-axis?
 (A) 3 (B) 4
 (C) -3 (D) 7
4. One exterior angle of a triangle at a vertex is 110° . If one of the interior opposite angles is 45° , what is the measure of the other interior opposite angle?
 (A) 55° (B) 65°
 (C) 75° (D) 85°
5. A notebook costs ₹5 more than a pen. If the total cost of one notebook and one pen is ₹45, what is the cost of the pen?
 (A) ₹20 (B) ₹25
 (C) ₹40 (D) ₹30
6. Two angles are equal and supplementary. Each angle measures
 (A) 45° (B) 60°
 (C) 90° (D) 120°
7. Match Column I with Column II.

Column I	Column II
a) Curved surface area of a cylinder	1) πr^2
b) Volume of a cylinder	2) $2\pi rh$
c) Area of base of a cylinder	3) $2\pi r(h+r)$
d) Total surface area of a cylinder	4) $\pi r^2 h$

(A) a-2, b-4, c-1, d-3
 (B) a-3, b-4, c-1, d-2
 (C) a-2, b-3, c-4, d-1
 (D) a-4, b-2, c-1, d-3

8. A shopkeeper divides ₹5 equally among 4 children. The amount received by each child is _____.
 (A) 1.25 (B) 1.5
 (C) 1.75 (D) 2
9. Which of the following is an axiom according to Euclid?
 (A) A straight line can be drawn joining any two points
 (B) A circle can be drawn with any centre and radius
 (C) Things equal to the same thing are equal to one another
 (D) All right angles are equal
10. In a pie chart, a sector has a central angle of 90° . What percentage of the total does it represent?
 (A) 15% (B) 20%
 (C) 25% (D) 30%
11. Which of the following is a solution of the equation $x + y = 10$?
 (A) (3, 6) (B) (4, 6)
 (C) (5, 4) (D) (2, 9)
12. Three sticks are of lengths 4 cm, 9 cm, and 14 cm. If a student tries to form a triangle using them, what will happen?
 (A) An acute triangle will be formed
 (B) An obtuse triangle will be formed
 (C) A right triangle will be formed
 (D) A triangle cannot be formed
13. Which of the following numbers has a terminating decimal expansion?
 (A) $\frac{3}{7}$ (B) $\frac{5}{8}$
 (C) $\frac{2}{9}$ (D) $\frac{7}{11}$
14. An angle measures 35° . How is it classified?
 (A) Zero angle (B) Acute angle
 (C) Obtuse angle (D) Reflex angle
15. Two lamp posts are placed at points (2, 0) and (8, 0) on a straight road. What is the distance between them?
 (A) 6 units (B) 10 units
 (C) 4 units (D) 8 units
16. A triangular park has two equal sides and all its angles are less than 90° . How is this triangle best described?
 (A) Scalene acute triangle
 (B) Isosceles acute triangle
 (C) Equilateral triangle
 (D) Right-angled triangle
17. A ladder reaches the top of a wall forming a right-angled triangle with the ground. If the ladder is 13 m long and the distance from the wall is 5 m, how high is the wall?
 (A) 8 m (B) 10 m
 (C) 12 m (D) 15 m
18. How many solutions does the equation $3x + 2y = 12$ have?
 (A) No solution
 (B) Exactly one solution
 (C) Exactly two solutions
 (D) Infinitely many solutions
19. The radius of a circle whose circumference is 44 cm is _____.
 (A) 14 cm (B) 7 cm
 (C) 22 cm (D) 4 cm

20. The points A $(-2, 0)$, B $(2, 0)$, and C $(0, 4)$ form a triangle. What type of triangle is formed?

- (A) Scalene
- (B) Isosceles
- (C) Equilateral
- (D) Right-angled

21. The difference between the areas of two circles is 462 cm^2 . If the radius of the smaller circle is 7 cm, the radius of the larger circle is _____.

- (A) 14 cm
- (B) 21 cm
- (C) 28 cm
- (D) 35 cm

22. Which of the following statements is true?

- (A) A line segment has no definite length
- (B) A ray has two endpoints
- (C) A line has no endpoints
- (D) A ray can be extended in both directions

23. The sum of two numbers is 50. One number is 6 more than the other. What is the larger number?

- (A) 22
- (B) 28
- (C) 44
- (D) 56

24. A number when multiplied by 4 and reduced by 9 gives 31. Find the number.

- (A) 5
- (B) 10
- (C) 8
- (D) 12

25. A student writes 0.375 as a fraction. Which is the correct fraction?

- (A) $\frac{3}{8}$
- (B) $\frac{5}{8}$
- (C) $\frac{15}{40}$
- (D) $\frac{375}{100}$

26. The polynomial $x^3 - 4x^2 - x + 4$ represents the volume of a solid. Which expression is a factor of the polynomial?

- (A) $x + 2$
- (B) $x - 2$
- (C) $x - 4$
- (D) $x + 4$

27. Two parallel lines are cut by a transversal. One interior angle on the same side is $3x + 10^\circ$ and the other is $5x - 30^\circ$. The value of x is _____.

- (A) 20
- (B) 25
- (C) 30
- (D) 35

28. A trapezium has its two parallel sides measuring 10 cm and 18 cm. A line joins the midpoints of the non-parallel sides. What is the length of this line?

- (A) 12 cm
- (B) 14 cm
- (C) 18 cm
- (D) 28 cm

29. If the mean of the following distribution is 6. Then the value of p is _____

X	2	4	6	8	P+5
F	3	2	3	1	2

- (A) 2
- (B) 7
- (C) 8
- (D) 5

30. Find Ten years ago, a person's age was half of what it will be after 10 years. What is the present age of the person?

- (A) 20 years
- (B) 25 years
- (C) 30 years
- (D) 40 years

31. A sector of a circle has radius 7 cm and angle 90° . What is the length of its arc?

- (A) 7 cm
- (B) 11 cm
- (C) 14 cm
- (D) 22 cm

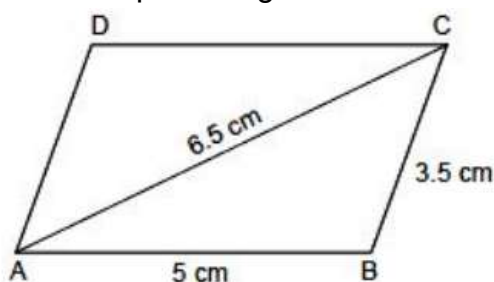
32. A student walks from point A (1, 2) to point B (4, 6) on a coordinate grid. What distance does the student cover?

- (A) 4 units (B) 6 units
(C) $\sqrt{13}$ units (D) $\sqrt{25}$ units

33. Which situation proves that the product of two irrational numbers can be rational?

- (A) $\sqrt{2} \times \sqrt{3}$ (B) $\sqrt{5} \times \sqrt{5}$
(C) $\sqrt{7} \times \sqrt{11}$ (D) $\sqrt{2} \times \sqrt{5}$

34. Two adjacent sides of a parallelogram measures 5 cm and 3.5 cm. One of its diagonal measures 6.5 cm. Find the area of the parallelogram.



- (A) $10\sqrt{2}cm^2$ (B) $10\sqrt{3}cm^2$
(C) $5\sqrt{3}cm^2$ (D) $4\sqrt{2}cm^2$

35. The perimeter of a rectangle is 60 cm. If the length is twice the breadth, what is the breadth?

- (A) 10 cm (B) 15 cm
(C) 20 cm (D) 30 cm

36. A bar graph is drawn using the scale 1 cm = 20 units. If a bar is 4 cm tall, what value does it represent?

- (A) 40 (B) 60
(C) 80 (D) 100

37. Two lines intersect forming four angles. If one angle is x° , its adjacent angle is $(2x + 30)^\circ$. Then the value of x is _____.

- (A) 30° (B) 40°
(C) 50° (D) 60°

38. The curved surface area of a cone is 550 cm^2 and its slant height is 25 cm. The radius of its base is _____.

- (A) 7 cm (B) 8 cm
(C) 9 cm (D) 11 cm

39. The midpoint of a line segment joining points A (2, -2) and B (6, 4) is _____.

- (A) (4, 1) (B) (2, 1)
(C) (8, 2) (D) (3, 0)

40. A cylindrical tank of radius 3.5 m and height 10 m is filled with water. How much water does it contain?

- (A) 275 m^3 (B) 350 m^3
(C) 385 m^3 (D) 550 m^3

41. **Statement-1:** The circumference of a circle is proportional to its radius.

Statement-2: The circumference of a circle is given by the formula $2\pi r$.

(A) Both Statement-1 and Statement-2 are true, and Statement-2 is the correct explanation of Statement-1.

(B) Both Statement-1 and Statement-2 are true, but Statement-2 is NOT the correct explanation of Statement-1.

(C) Statement-1 is true and Statement-2 is false.

(D) Statement-1 is false and Statement-2 is true.

42. Which of the following fractions will have a non-terminating repeating decimal expansion?

- (A) $7/20$ (B) $9/25$
(C) $11/30$ (D) $13/40$

43. Which of the following expressions can represent the area of a rectangular garden whose length is $x + 2$ m and breadth is x m?

- (A) $x^2 + 2x$ (B) $x^2 + \frac{2}{x}$
(C) $\frac{x^2}{2}$ (D) $x^{-2} + 2$

44. The area of a circular field is 154 sq. m. What is the radius of the field?

- (A) 14 m (B) 7 m
(C) 21 m (D) 11 m

45. The sides of a triangular garden are 8 m, 10 m, and 12 m. If fencing is to be done around the garden, what total length of fencing is required?

- (A) 20 m (B) 24 m
(C) 30 m (D) 36 m

46. A point P divides the line segment joining A (2, -2) and B (8, 4) internally in the ratio 1 : 2. What are the coordinates of P?

- (A) (4, 0) (B) (6, 2)
(C) (3, -1) (D) (5, 1)

47. In a parallelogram, one angle is measured as 65° . What is the measure of its adjacent angle?

- (A) 65° (B) 115°
(C) 125° (D) 180°

48. Two lines intersect. One angle is $(3x - 10)^\circ$ and its vertically opposite angle is $(2x + 20)^\circ$. Find the value of x .

- (A) 20 (B) 25
(C) 30 (D) 35

49. The polynomial representing the area of a plot is $x^2 - 9$. Which value of x makes the area zero?

- (A) 1 (B) 3
(C) 5 (D) 9

50. In a parallelogram, the diagonals intersect at a point O. If one diagonal is divided into two segments of 6 cm each, what is the length of that diagonal?

- (A) 6 cm (B) 9 cm
(C) 12 cm (D) 18 cm

KEY

- | | | | | | | |
|-------|-------|-------|-------|-------|-------|-------|
| 1) C | 2) B | 3) A | 4) B | 5) A | 6) C | 7) A |
| 8) A | 9) C | 10) C | 11) B | 12) D | 13) B | 14) B |
| 15) A | 16) B | 17) C | 18) D | 19) B | 20) B | 21) A |
| 22) C | 23) B | 24) B | 25) A | 26) C | 27) B | 28) B |
| 29) C | 30) C | 31) B | 32) D | 33) B | 34) B | 35) A |
| 36) C | 37) C | 38) A | 39) A | 40) C | 41) A | 42) C |
| 43) A | 44) B | 45) C | 46) A | 47) B | 48) C | 49) B |
| 50) C | | | | | | |